Solution: Basic Sum

Problem Statement

Calculate the Sum of the First `N` Natural Numbers Using a Recursive Approach.

The sum of first N natural numbers is equal to N + (N-1) + (N-2) + ... + (3) + (2) + (1). The following table shows a sample input/output description table:

Input (s)	Output (s)	Explanation
N = 5	Sum = 15	The first 5 natural numbers are 1, 2, 3, 4, and 5. The sum of these numbers is 1 + 2 + 3 + 4 + 5 = 15.
N = 10	Sum = 55	The first 10 natural numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10. The sum of these numbers is 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55.
N = 1	Sum = 1	The first natural number is 1. The sum of this number is 1.

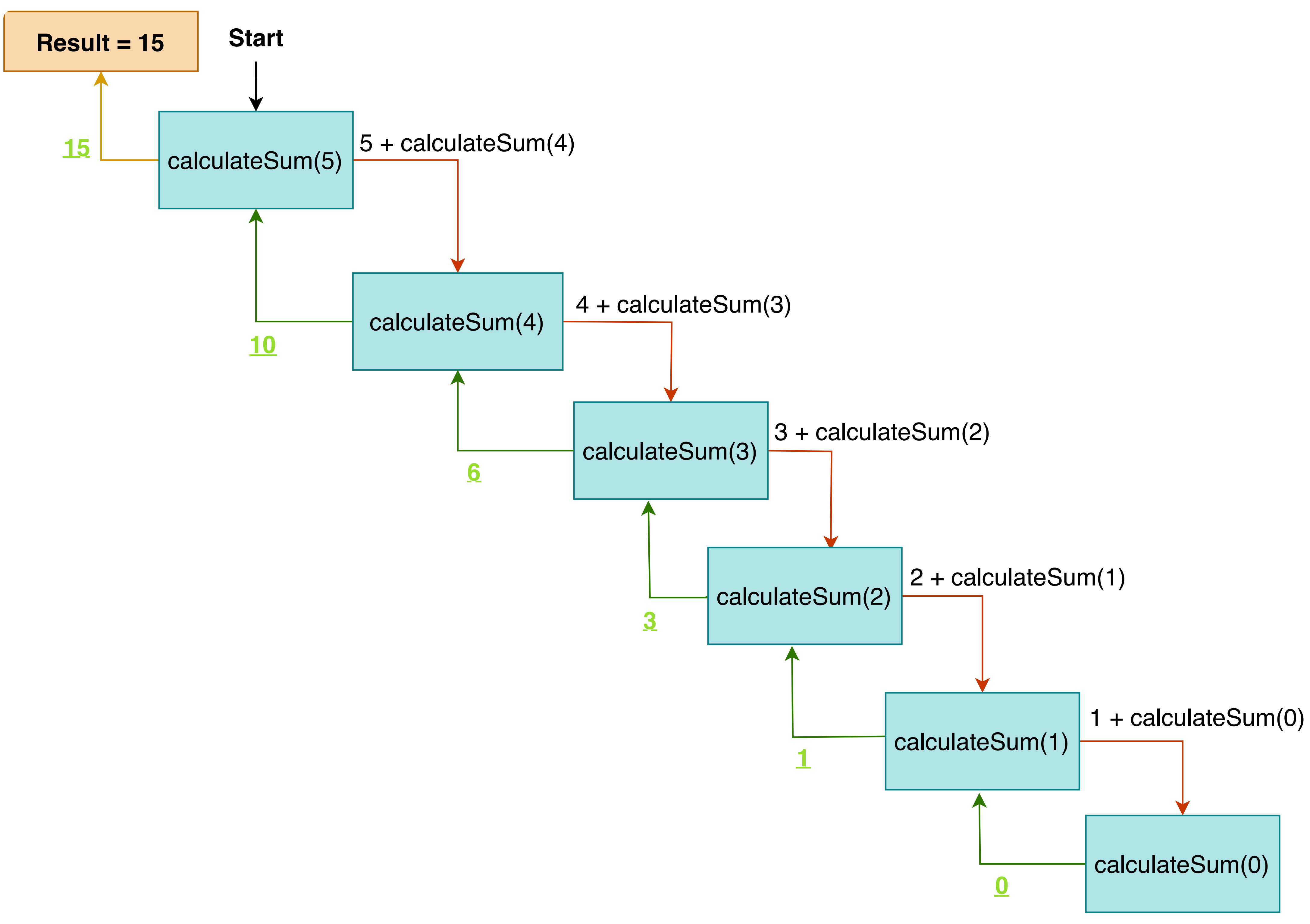
Solution

The algorithm to calculate the sum of the first N natural numbers using recursion can be defined as follows:

1. If `N` is equal to `0` or less, return `0` (base case).
2. Otherwise, return the sum of number `N` and the recursive call to calculate the sum of `N-1`.

The recursive approach works by breaking down the problem into smaller subproblems. We calculate the sum of N by adding N to the sum of the numbers from 1 to N-1. This process continues until the base case is reached, where N is 0 or less.

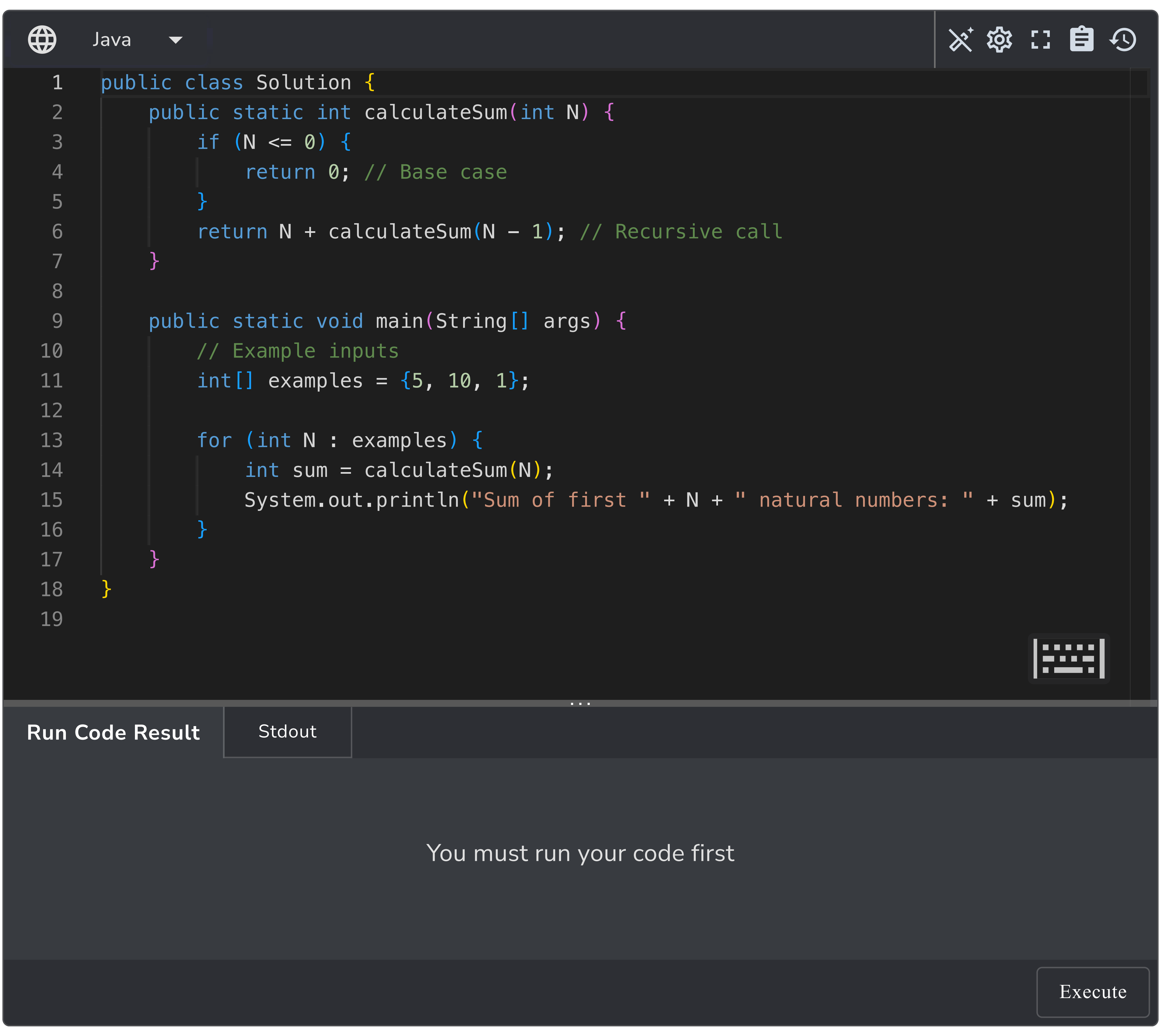
The base case stops the recursion and returns 0. By summing the numbers in a decreasing order, we ensure that each number is added to the previous sum, eventually giving us the total sum of the first N natural numbers.



Example dry run to calculate Sum of first 5 Natural numbers

Code

Here is the code for this algorithm:



```java
public class Solution {
    public static int calculateSum(int N) {
        if (N <= 0) {
            return 0; // Base case
        }
        return N + calculateSum(N - 1); // Recursive call
    }

    public static void main(String[] args) {
        // Example inputs
        int[] examples = {5, 10, 1};

        for (int N : examples) {
            int sum = calculateSum(N);
            System.out.println("Sum of first " + N + " natural numbers: " + sum);
        }
    }
}

```

You must run your code first

Time and Space Complexity

The time complexity of the algorithm is $O(N)$ since we make N recursive calls, where N is the input natural number.

The space complexity of the algorithm is also $O(N)$ because the maximum depth of the recursion is N. Each recursive call adds a new stack frame, consuming additional space on the call stack.

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